PROJECT 9 — MILESTONE 1 · TECHNICAL REPORT

Intelligent Property Price Prediction

ML-Based Valuation System — Washington State Housing Market

BEST MODEL	R^2 SCORE	DATASET
Linear Regression	**0.8028**	**4,458 records**

TEAM MEMBERS

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SECTION 01

Problem Statement & Dataset

Residential property pricing is one of the most consequential decisions in personal finance — yet the process remains surprisingly opaque. Buyers, sellers, and investors often rely on manual appraisals that are subjective, slow, and inconsistent across evaluators. Two properties on the same street can differ by hundreds of thousands of dollars based on factors that aren't always obvious from a listing: interior square footage versus lot size, renovation history, floor level, proximity to water.

Our objective was to replace intuition with evidence. We framed this as a supervised regression problem: given a set of measurable property attributes, can we build a model that reliably estimates what a home will actually sell for? And beyond the prediction itself — can we explain why certain features matter more than others?

> *"The goal wasn't just to predict a price — it was to build something that could tell an investor which features actually drive value, and by how much."*

📐 Structural — sqft, rooms, floors
📍 Geographic — city, postal zone
🏗️ Condition — rating, view, waterfront
📅 Temporal — build year, sale season

DATASET OVERVIEW

We used residential transaction records from the greater Seattle metro area, covering a three-month window in 2014 across 44 municipalities and 77 postal zones. After removing zero-price records and entries with invalid room counts, our working dataset comprised 4,458 clean transactions.

4,600	4,458	18	135
RAW RECORDS	AFTER CLEANING	RAW FEATURES	AFTER ENGINEERING

FEATURE REFERENCE

Feature	Type	Description
sqft_living	Numerical	Interior living area in square feet
sqft_lot	Numerical	Total lot size in square feet
bedrooms / bathrooms	Numerical	Room counts
floors	Numerical	Number of floor levels
waterfront / view	Binary / Ordinal	Waterfront flag, view quality rating 0–4
condition	Ordinal	Property condition rating 1–5
sqft_above / sqft_basement	Numerical	Above grade and basement area
yr_built / yr_renovated	Numerical	Construction and renovation year
city / statezip	Categorical	Municipality and postal zone

SECTION 02

Exploratory Data Analysis

Before writing a single line of model code, we spent time understanding what the data was actually telling us. EDA shaped several of our most impactful design decisions — including the log transform on price, which ended up being the single biggest improvement to model performance.

PRICE DISTRIBUTION

The raw price distribution was heavily right-skewed, with a long tail of high-value luxury properties pulling the mean well above the median. Most transactions clustered between $200K–$600K, but a handful of outliers exceeded $3M. This skew is problematic for linear models because it causes them to over-weight the expensive end of the market and under-perform on typical mid-range properties. Applying `np.log1p()` to the price target before training produced a near-normal distribution, allowing the model to treat percentage price errors equally regardless of price level.

> *Key finding: log-transforming price pushed R^2 from 0.733 → 0.8028 — a bigger gain than adding XGBoost or tuning Random Forest.*

CORRELATION ANALYSIS

Among all numerical features, sqft_living showed the strongest positive correlation with price (r ≈ 0.70), followed by sqft_above (r ≈ 0.61) and bathrooms (r ≈ 0.53). These three features alone explain a significant portion of price variance before any encoding or scaling.
Notably, sqft_lot had a much weaker correlation (r ≈ 0.09), suggesting that land size matters less than usable interior space in this urban-dense market — an insight that shaped our feature engineering decisions.

GEOGRAPHIC PATTERNS

Plotting average sale price by city revealed sharp geographic stratification. Properties in Medina, Mercer Island, and Bellevue commanded median prices 2–3× higher than comparable-sized homes in Auburn or Federal Way. This confirmed that location encoding would be critical — not just as a minor adjustment, but as a primary pricing signal.

KEY EDA FINDINGS

📊 **Price is right-skewed, not normally distributed**

Log transformation was necessary before training — it normalizes residuals and improves model convergence.

📐 **Living area is the strongest numerical predictor**

sqft_living correlates with price at r ≈ 0.70, far above any other single feature.

📍 **Geographic location creates significant price tiers**

City and ZIP code account for ~14% of model importance after one-hot encoding — nearly as predictive as structural features combined.

🏗️ **Waterfront is a binary premium, not a gradient**

Waterfront properties sell at a significant premium regardless of other attributes — it functions almost like a category switch.

🔢 **Lot size matters much less than interior size**

sqft_lot had r ≈ 0.09 with price — useful context for our living_lot_ratio feature but not independently strong.

Methodology

PREPROCESSING PIPELINE

We built a modular, reusable preprocessing pipeline in `preprocessing.py` that handles the full transformation sequence in a fixed, reproducible order. Every transformation is fitted on training data only and applied to test data — this is the key to preventing data leakage.

1. **Drop non-informative columns**
   Removed*street*and*country*—these add noise without predictive value in a single-region dataset.

2. **Date parsing & temporal feature extraction**
   Extracted*year_sold*and*month_sold*fromthetransaction date to capture seasonality effects.

3. **Feature engineering (4 new features)**
   total_sqft,bed_bath_ratio,living_lot_ratio,is_renovated — each encodes a relationship not visible in raw columns.

4. **Invalid record removal**
   Dropped49zero-pricerecords and 2 entries with zero bedrooms or bathrooms — these represent data entry errors, not real transactions.

5. **Dual outlier treatment**
   Quantiletrim(1st–99thpct) on price to remove extreme outliers; IQR clipping on all feature columns to preserve row count.

6. **Train/test split (80/20) — before encoding**
   SplithappensbeforeOHEandscalingtoprevent test set information leaking into the training transformers.

7. **One-hotencoding (city + statezip)**
   `pd.get_dummies` withdrop_first=True. Train set defines the schema — test columns are aligned, missing ones filled with 0.

8. **StandardScaler on continuous features**
   Fittedontrainingsetonly,appliedtoboth.Covers sqft columns, room counts, age, and temporal features.

9. **Logtransformonprice target**
   `np.log1p(price)` normalizesthe right-skewed distribution. Reversed with `np.expm1()` at inference time.

MODELS TRAINED

We trained five models, selected to cover the spectrum from simple linear to ensemble nonlinear methods. All models were auto-evaluated using the same test set — no manual winner selection.

Model	Key Parameters	Rationale
Linear Regression	OLS, no regularization	Baseline interpretable model — expected to perform well on linear data
Ridge Regression	alpha=10	Regularized linear — better for high-dimensional sparse OHE features
Random Forest	n_estimators=200, random_state=42	Ensemble tree — captures nonlinear interactions
Decision Tree	max_depth=6, min_samples_split=10	Single tree baseline — interpretable but prone to overfitting
XGBoost	n_estimators=500, lr=0.05, max_depth=6	Gradient boosting — state-of-the-art for tabular regression

SECTIONS 04 & 05

Evaluation Metrics & Optimisation

Each model was evaluated on the held-out test set (20% of data, 892 records) using three complementary metrics. We also ran 5-fold cross-validation on the training set to confirm that results were stable and not artifacts of a fortunate split.

Model	R^2 Score	CV R^2 (5-fold)	MAE	RMSE
✅ **Linear Regression**	**0.8028**	**0.8160**	**$81,492**	**$138,943**
Ridge Regression	0.7906	0.8083	$84,840	$143,394
XGBoost	0.7883	0.8056	$85,287	$142,682
Random Forest	0.7346	0.7572	$96,655	$159,300
Decision Tree	0.5193	0.5619	$136,503	$204,039

Linear Regression won because after log-transform and OHE (135 features, mostly sparse binary), the price signal is already additive-linear. CV R^2=0.8160 > Test R^2=0.8028 confirms zero overfitting.

FEATURE IMPORTANCE

Extracted from the Random Forest (which provides importances regardless of model selection):

Rank	Feature	Importance	What It Means
1	sqft_living	44.2%	Interior living area — dominant price signal by a wide margin
2	total_sqft	5.5%	Engineered feature combining living + basement area
3	city_Seattle	5.2%	Seattle location premium — significant geographic uplift
4	sqft_lot	4.1%	Lot size — secondary to interior area but still relevant
5	property_age	3.8%	Engineered from yr_built — newer homes command premium

SECTION 05

Optimisation Steps

We made five deliberate improvements over the base notebook — each documented with its measured impact:

✅ **Log transform on price target** +7.0% R^2
Applied np.log1p() before training. Normalized the right-skewed price distribution. Pushed R^2 from 0.733 → 0.8028. Single biggest improvement in the project.

✅ **Feature engineering — 4 new columns** +3.5% R^2
Created total_sqft, bed_bath_ratio, living_lot_ratio, is_renovated. These encode relationships that raw columns don't capture individually.

✅ **Dual outlier treatment strategy**
Quantile trim (1st–99th pct) on price removes extreme transactions. IQR clipping on feature columns handles outliers without losing rows.

✅ **Data leakage prevention**
Train/test split happens before OHE and scaling. Transformers fitted on train only, applied to both. Standard practice but often skipped in basic notebooks.

✅ **5-fold cross-validation across all 5 models**
Added CV R^2 column to results table. Confirms model stability — Linear Regression CV R^2 of 0.8160 is higher than test R^2, ruling out lucky splits.

SECTION 06

Team Contributions & Conclusion

Member	Role	Specific Contributions
Vedant Madne 2401010499 · Lead	Model Training & Evaluation	Trained all 5 models, implemented 5-fold CV, auto-selection logic, feature importance analysis, results table, model selection narrative
Aryu Rao 2401010107	Data & Preprocessing	Data sourcing, built modular preprocessing pipeline, feature engineering (4 new features), outlier treatment strategy, log transform implementation
Vidhi Singhal 2401020075	UI & Deployment	Streamlit web application, inference pipeline (predict_property function), Streamlit Cloud deployment, live URL, UI/UX design
Shitanshu Tiwari 2401010440	Notebook & Documentation	Jupyter notebook with EDA, visualizations, written analysis, project report, README, GitHub repository management

CONCLUSION

We set out to answer a simple question and ended up building something we're genuinely proud of. The final system — five models trained on 4,458 cleaned transactions, deployed as a live web app — predicts Seattle-area residential sale prices with $R^2 = 0.8028$ and an average error of $81,492.

The most important lesson from this project wasn't about model selection — it was about data preparation. The log transform on price, the four engineered features, and the strict data leakage controls collectively contributed more to model performance than any algorithm choice. Linear Regression, which most people dismiss as "too simple," outperformed XGBoost and Random Forest because the data, once properly prepared, exhibited strongly linear structure.

For future work, the clearest improvement paths are incorporating more recent transaction data (the 2014 dataset can't reflect post-2020 market shifts), adding school district ratings and walkability scores as neighbourhood quality signals, and wrapping the model in a REST API for programmatic access by downstream applications.

⚠ All price predictions are statistical estimates based on 2014 transaction data. This system is intended for educational purposes and does not constitute professional real estate or financial advice.

0.8028	$81K	5	135
FINAL R^2	MEAN ABS. ERROR	MODELS COMPARED	FEATURES USED